		$\approx 4.2 \times 10$ J
14	**D**	Periods of waves X and Y each occupy 3 large squares on the time axis. Difference between waves X and Y is 1 large square along the time axis. Hence, phase difference = 120° Ratio $= \dfrac{\text{intensity of wave X}}{\text{intensity of wave Y}}$ (intensity $\propto$ amplitude^2) $= \left(\dfrac{\text{15 small squares}}{\text{10 small squares}}\right)^2$ $= 2.25$
15	**B**	After first filter, intensity of light = I